

BUSINESS ANALYTICS

DESCRIPTIVE ANALYSIS

Amazon E-commerce
Data

AGENDA

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METHODOLOGY

Development Environment

- **GOOGLE COLAB** (CLOUD-BASED PYTHON NOTEBOOK)
- **GOOGLE DRIVE** INTEGRATION FOR DATA STORAGE & ACCESS

Python Data Analysis

- **PANDAS** (DATA LOADING, CLEANING, MERGING & AGGREGATION)
- **NumPy** (Numerical operations & category mapping)
- **Matplotlib & Seaborn** (Statistical analysis and visualizations)

Interactive Dashboard

- **HTML, CSS & JAVASCRIPT**
- **Chart.js** for interactive charts and KPIs
- **IPython.display** for rendering dashboard components

INTRODUCTION – DATASET

Open e-commerce 1.0: Five years of crowdsourced U.S. Amazon purchase histories with user demographics¹



amazon-
purchases.csv
ORDER, SALES INFORMATION
CATEGORIES



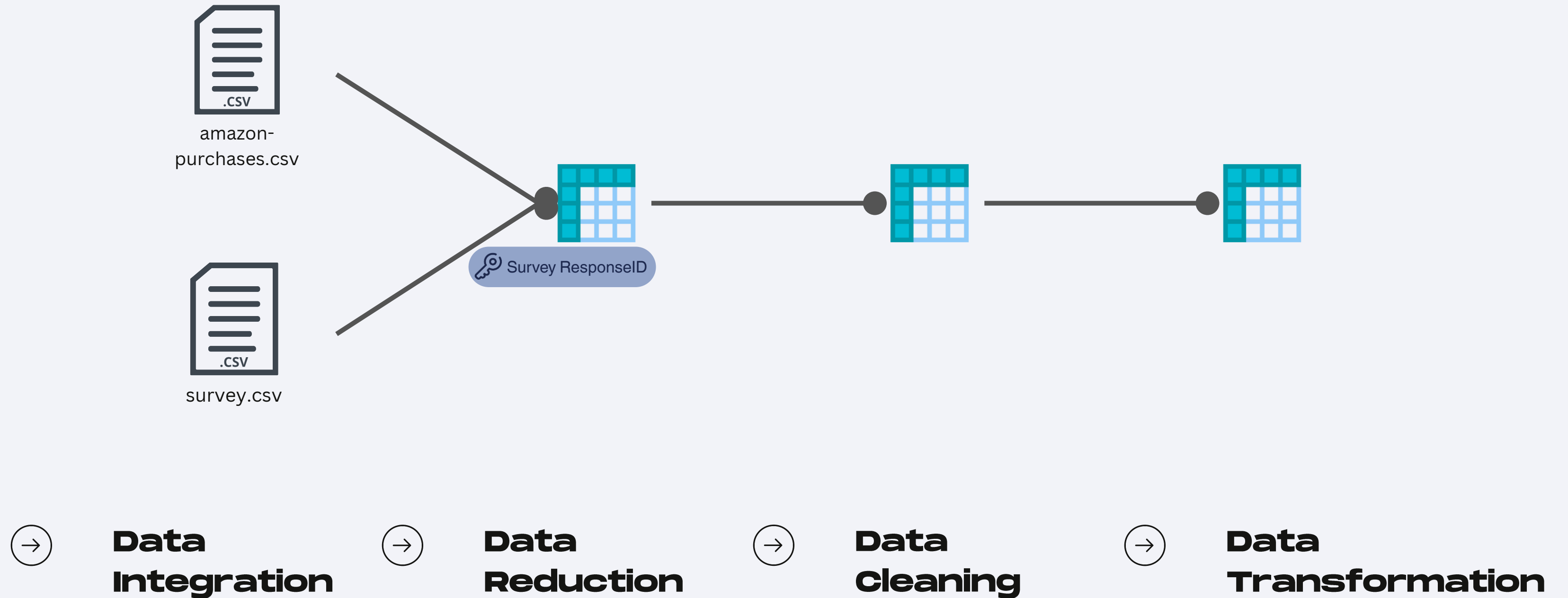
survey.csv
DEMOGRAPHIC DATA
CONSUMER LEVEL
VARIABLES

¹Berke, A., Calacci, D., Mahari, R., Yabe, T., Larson, K., Pentland, S., 2023. Open e-commerce 1.0: Five years of crowdsourced U.S. Amazon purchase histories with user demographics. <https://doi.org/10.7910/DVN/YGLYDY>

BUSINESS PROBLEM

1. How do demographic factors define the highest-value customers
2. Which geographic locations (US state level) drive customer value the most
3. Which top-selling products can be identified

DATA INTEGRATION AND PREPROCESSING



PRELIMINARY ANALYSIS

- 1,761,259 Rows
- 9 Category Datapoints
- 6 Numerical Datapoints
- Time Range: Jan. 01 2018 - Aug. 15 2024
- Total Revenue: \$41,730,373
- Avg. Order Value: \$24

PRELIMINARY ANALYSIS

Category	(US) State	(Product) Category	Agg_Category	Age	Gender	Income	Education
Unique	52	1871	21	6	4	7	5
Top	California	ABIS_BOOK	Other / Misc	25 - 34 years	Female	\$100,000 - \$149,999	Bachelor's degree

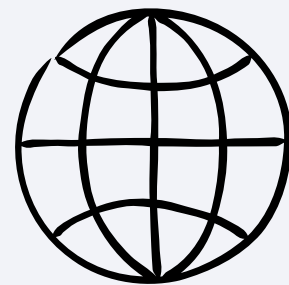
EXPLORATORY DATA ANALYSIS



TOTAL PRICE DISTRIBUTION
OUTLIERS



CATEGORIES BY PURCHASE
COUNT



DEMOGRAPHIC PATTERNS
1.REVENUE BY INCOME GROUP
2.REVENUE BY AGE GROUP
3.REVENUE BY STATE



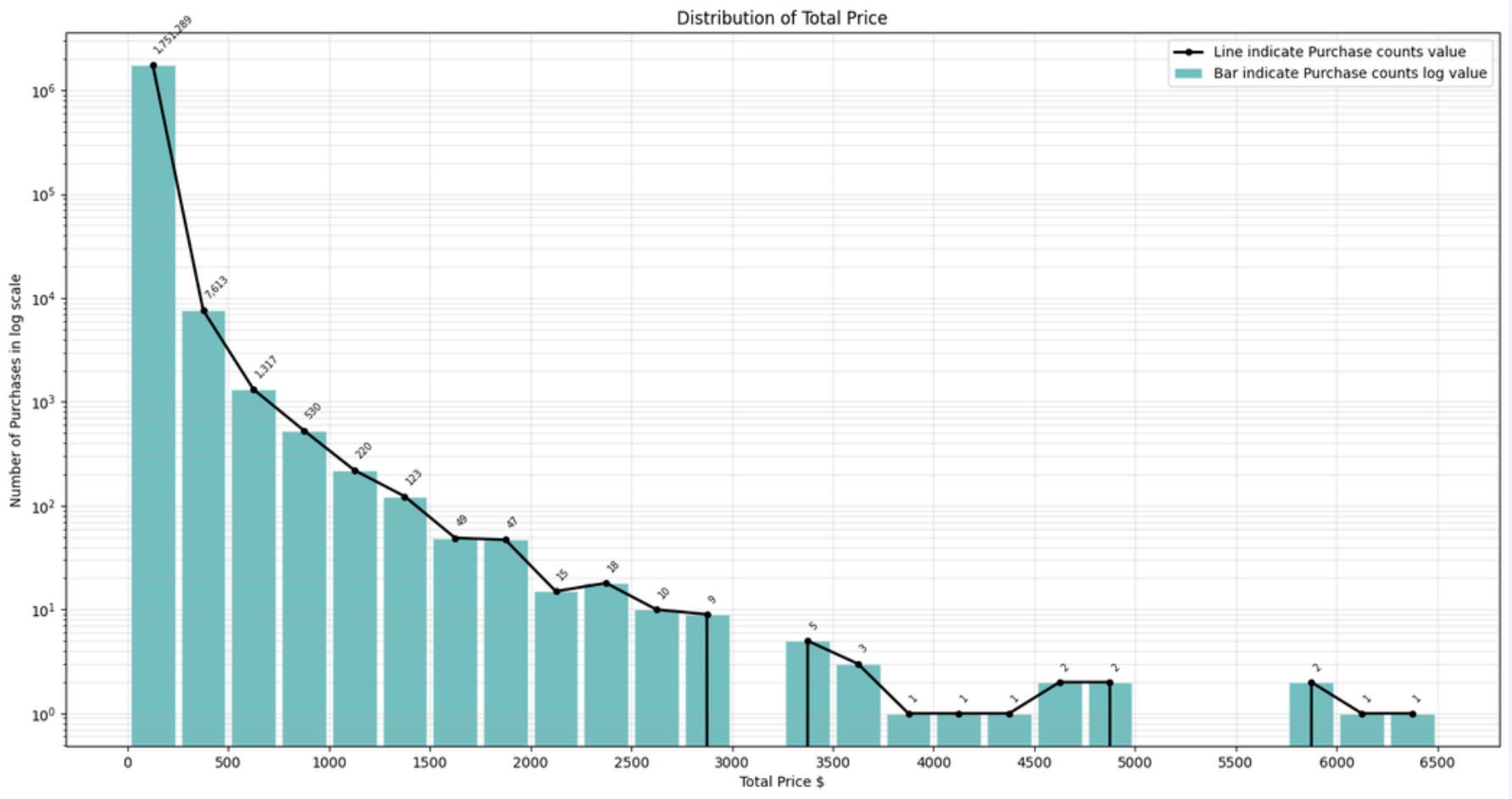
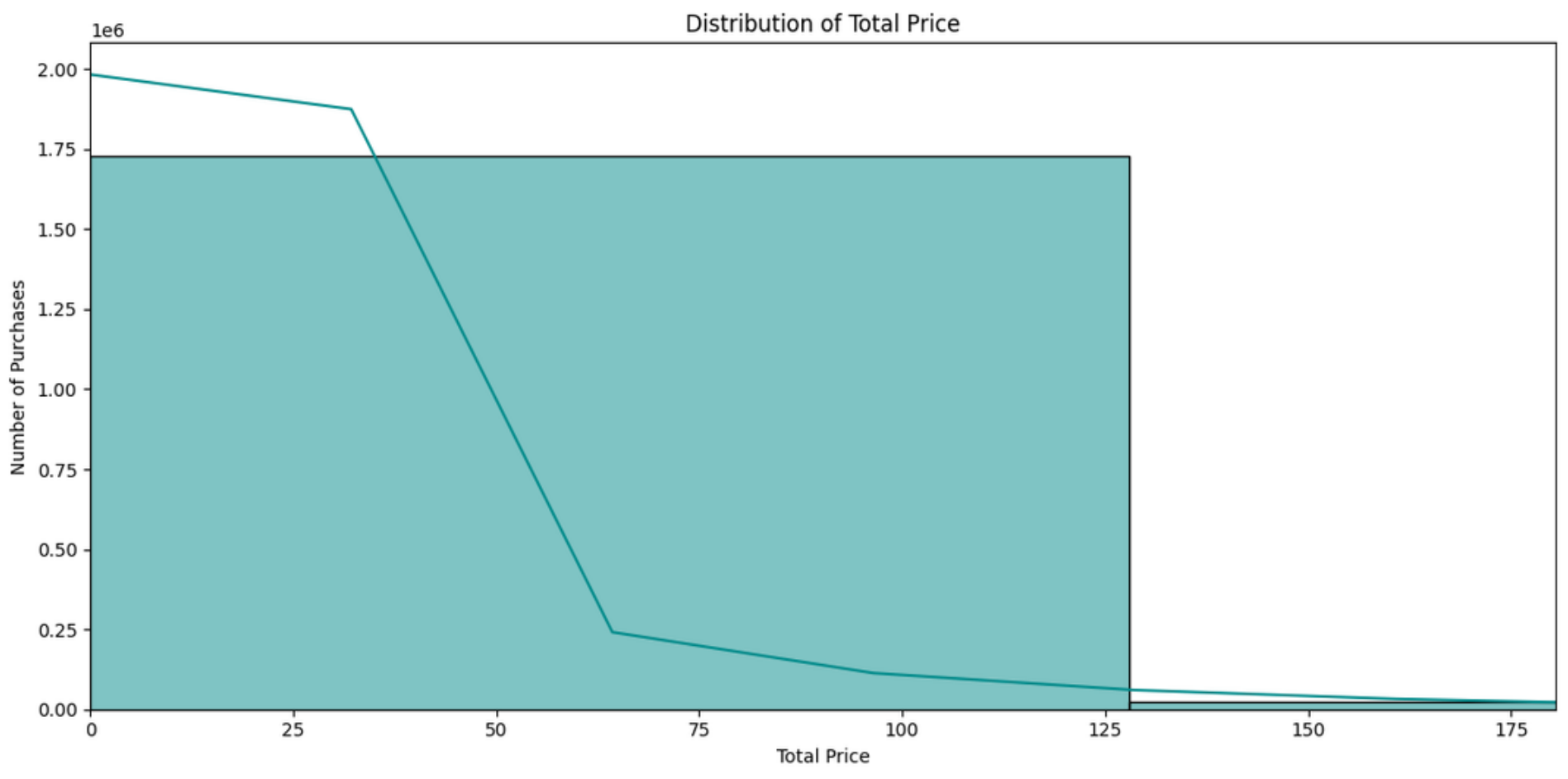
TIME PATTERNS
1.MONTHLY TREND
2.DAYS IN WEEK PATTERN

HOW OUR METRICS WORK

- Total Price is calculated per row, as unit price times quantity, computed once per purchased item.
- Total Revenue is a single sum for the entire dataset across all 1.76 million rows.
- The histogram shows the spread of individual purchase values, not the single total revenue figure.
- For the demographic and geographic charts, we just group those same purchase amounts by the buyer's income, age, or state label that came from the survey, and sum within each group.
- All our revenue and count metrics are computed from the Amazon purchase transaction data: Total Price, categories, dates.
- The survey supplies the demographic labels we use to slice and group those purchases.

TOTAL PRICE DISTRIBUTION

SHAPE OF PURCHASE VALUES (TOTAL PRICE)



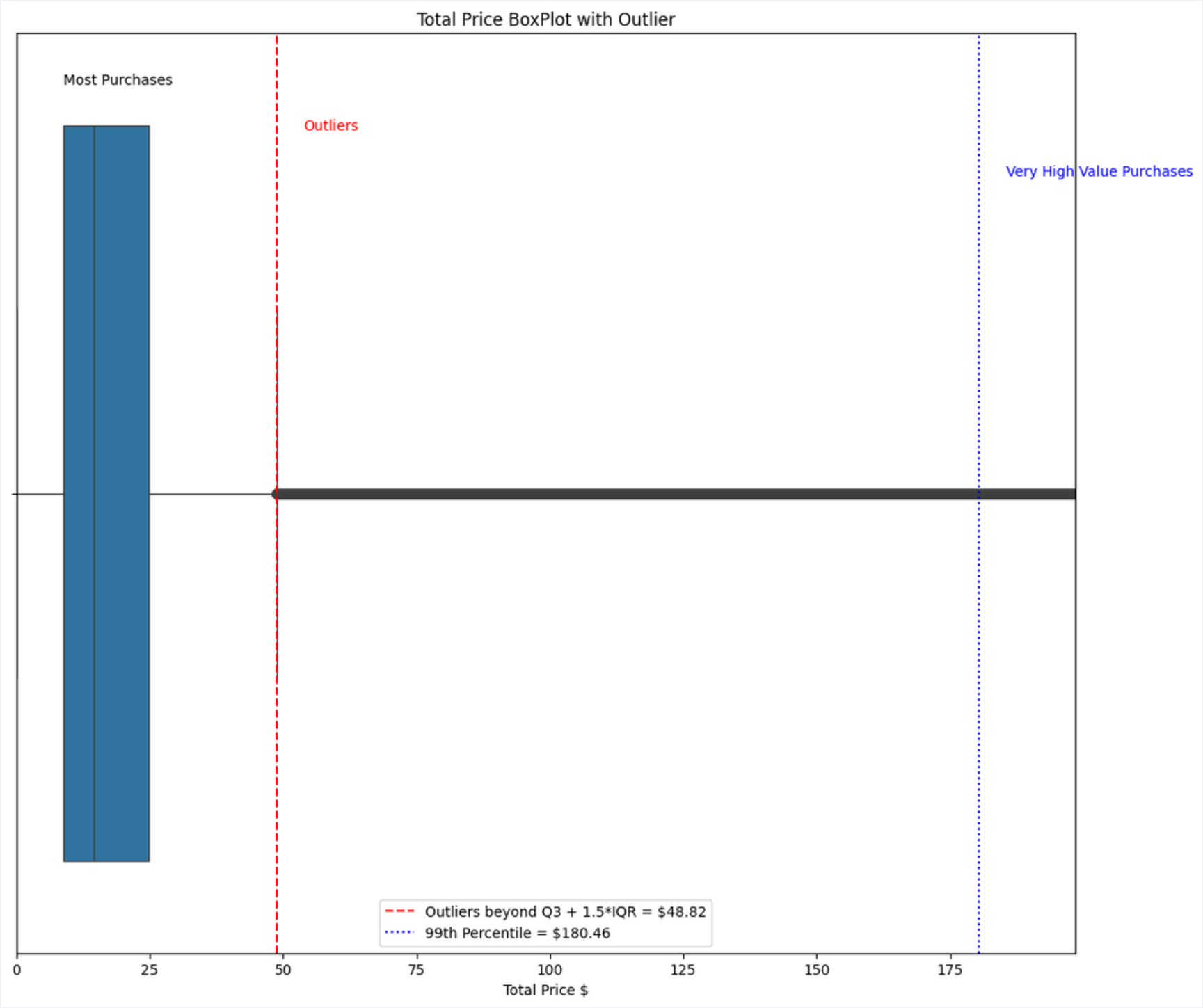
Total Revenue ≈ \$41.7M

Median purchase value: \$14.49 ; Mean: \$23.69 ; 75th Percentile: \$24.84

Maximum single line item: \$6,398.95

Few very expensive purchases create a long right tail; Most transactions are low value, with revenue highly skewed.

OUTLIERS



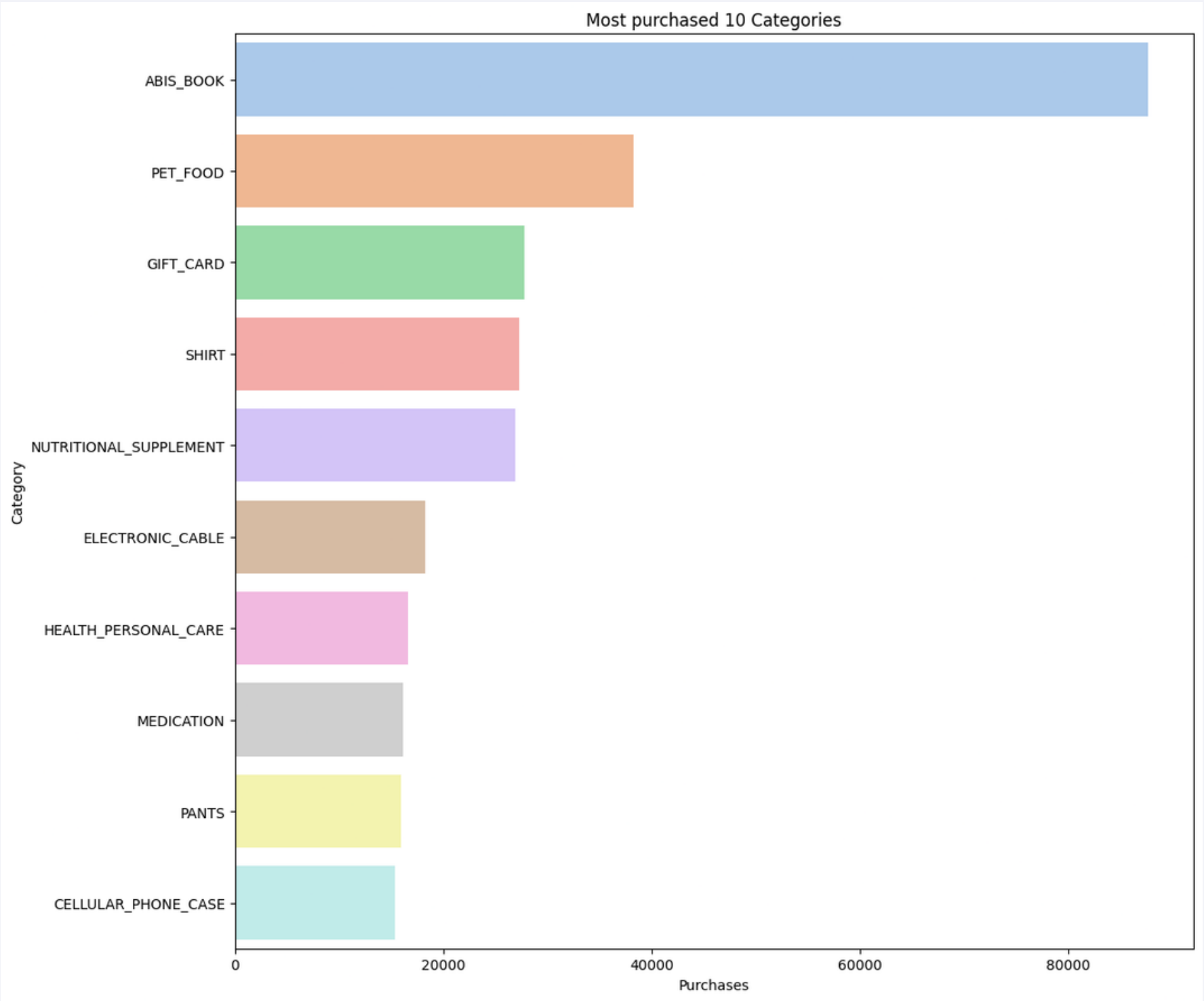
Right-Skewed Distribution: A few large ones pull the mean to the right

There are many small values and few large ones

Outliers: Purchases more than \$48.82 ($Q3 + 1.5 * IQR$) are outliers, significantly different from the majority

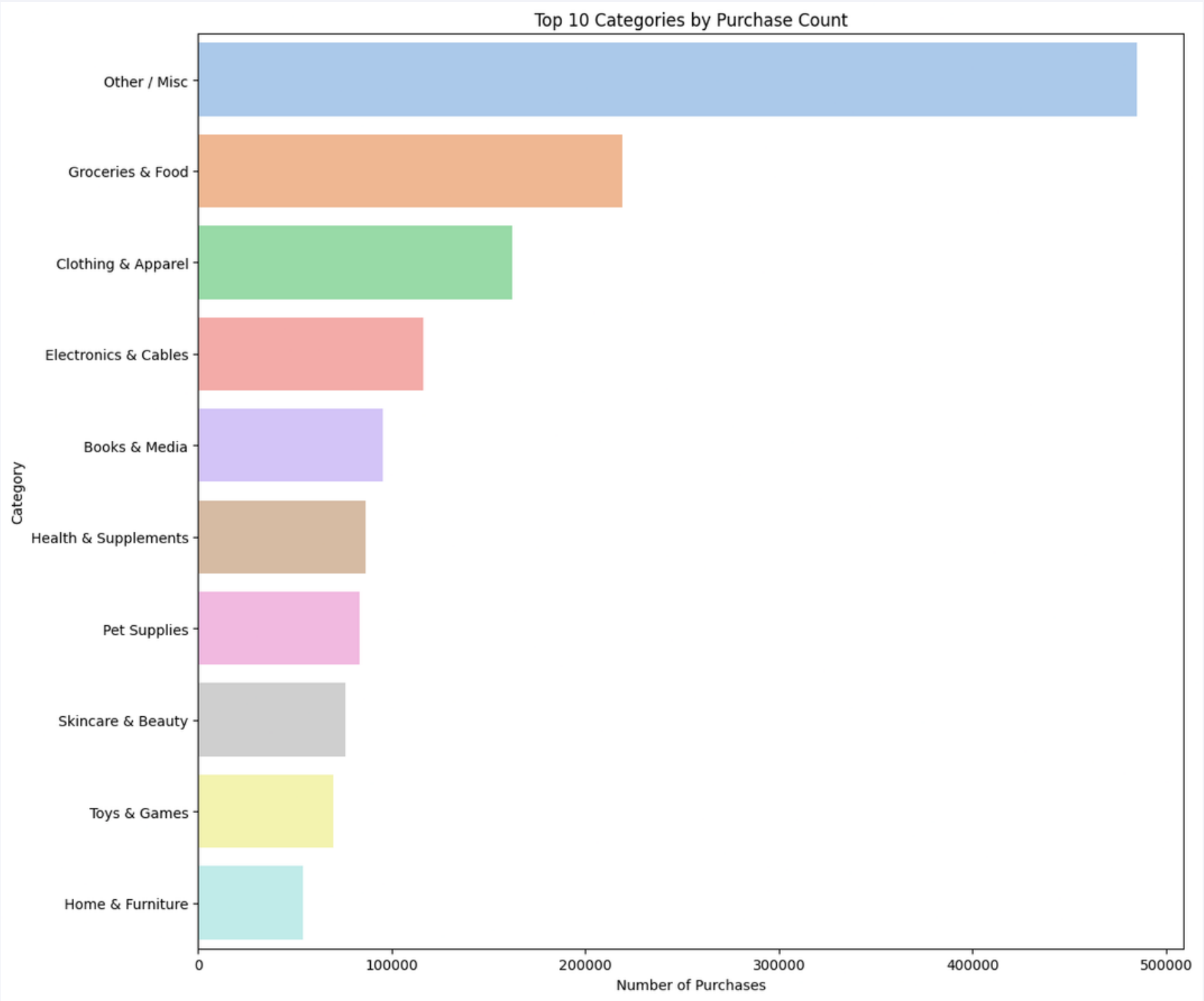
TOP CATEGORIES (BY PURCHASES)

Default Categories



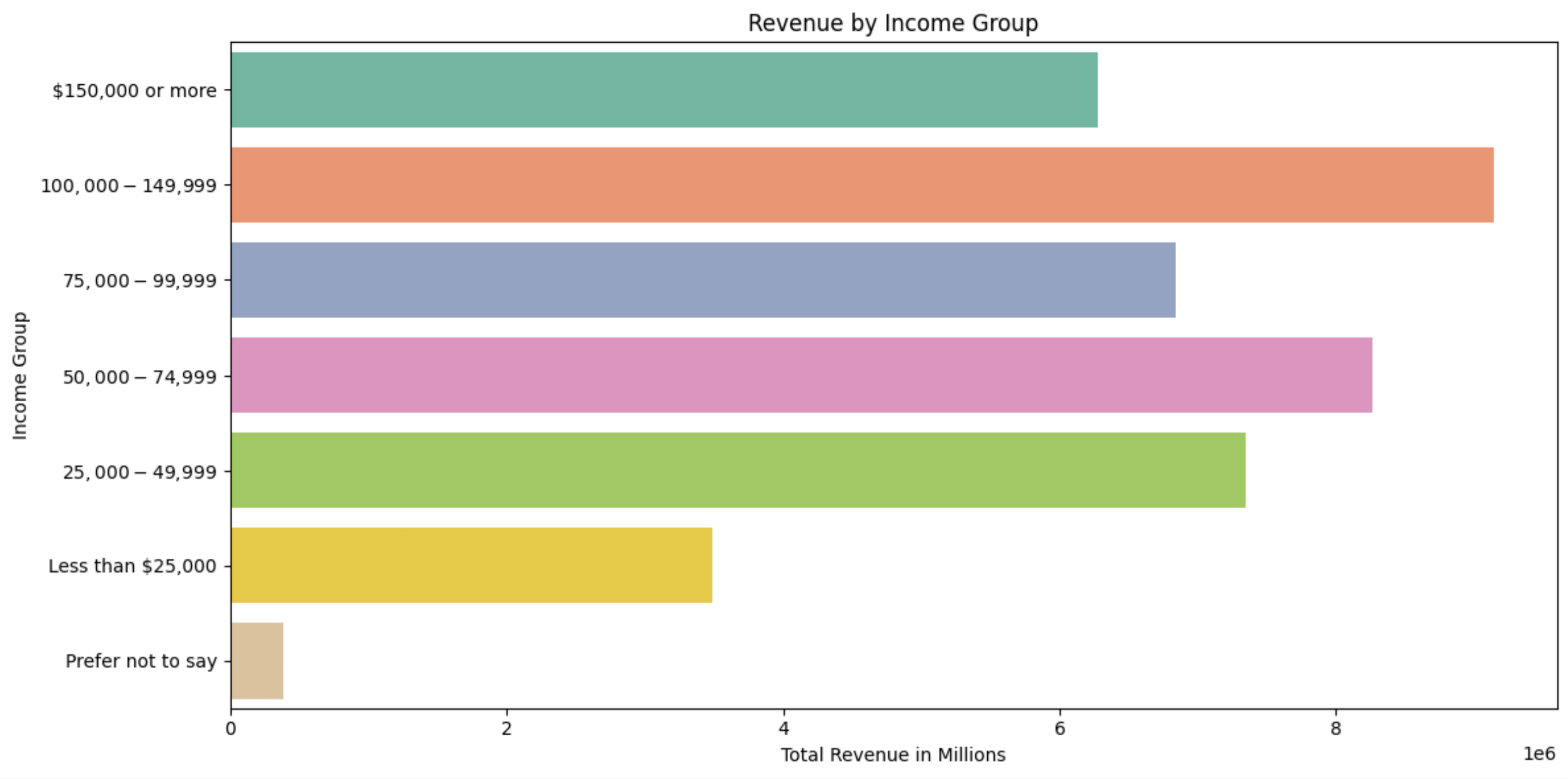
Top raw category: abis_book with 87,619 purchases
Others: pet_food, gift_card, shirt

Aggregated Categories



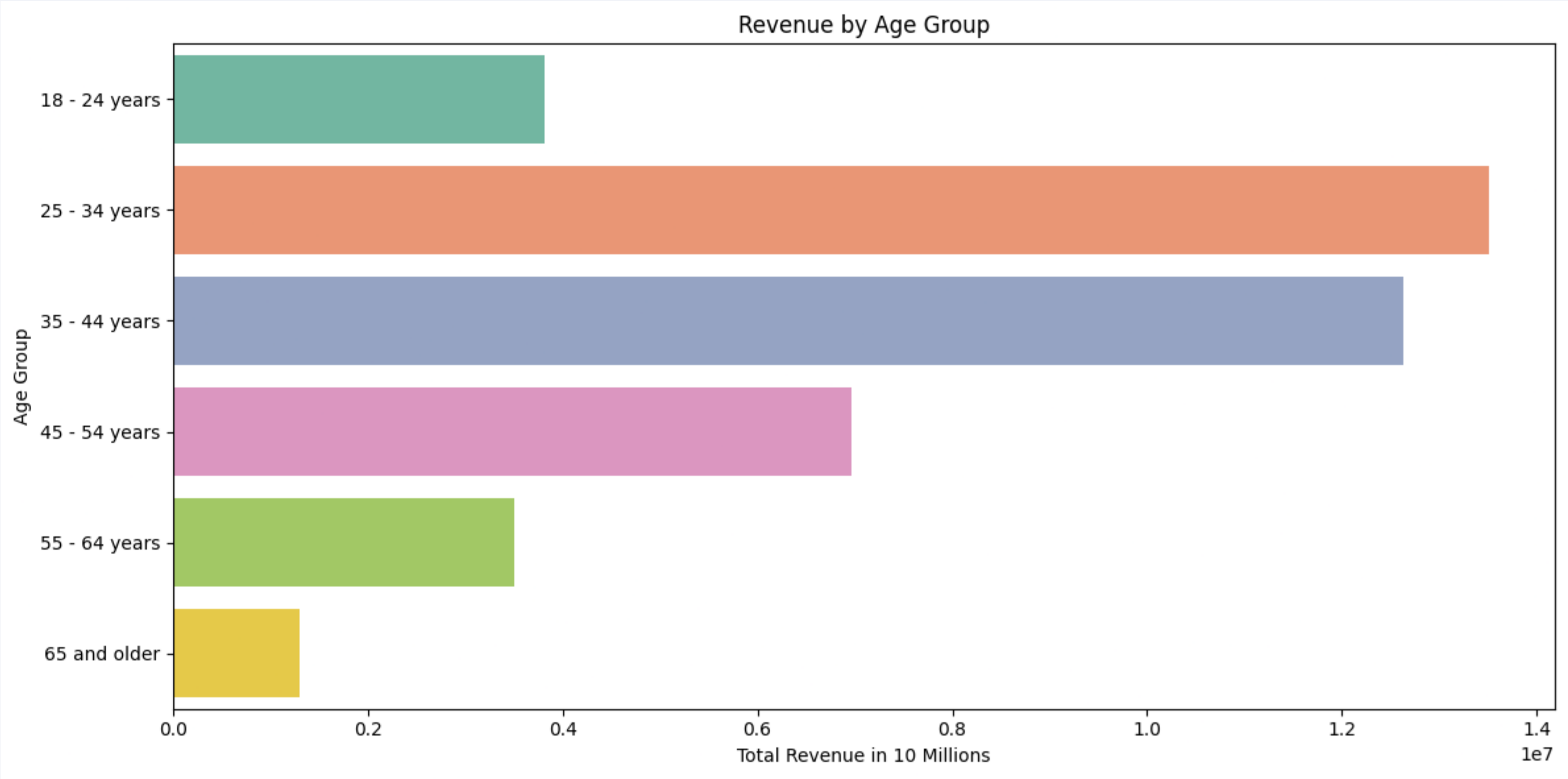
Top aggregate families: groceries & food with 219,287 purchases
Top categories in Other / Misc with purchases:
electronic_cable -18268, physical_movie -6920, shorts -6274

REVENUE BY INCOME GROUP



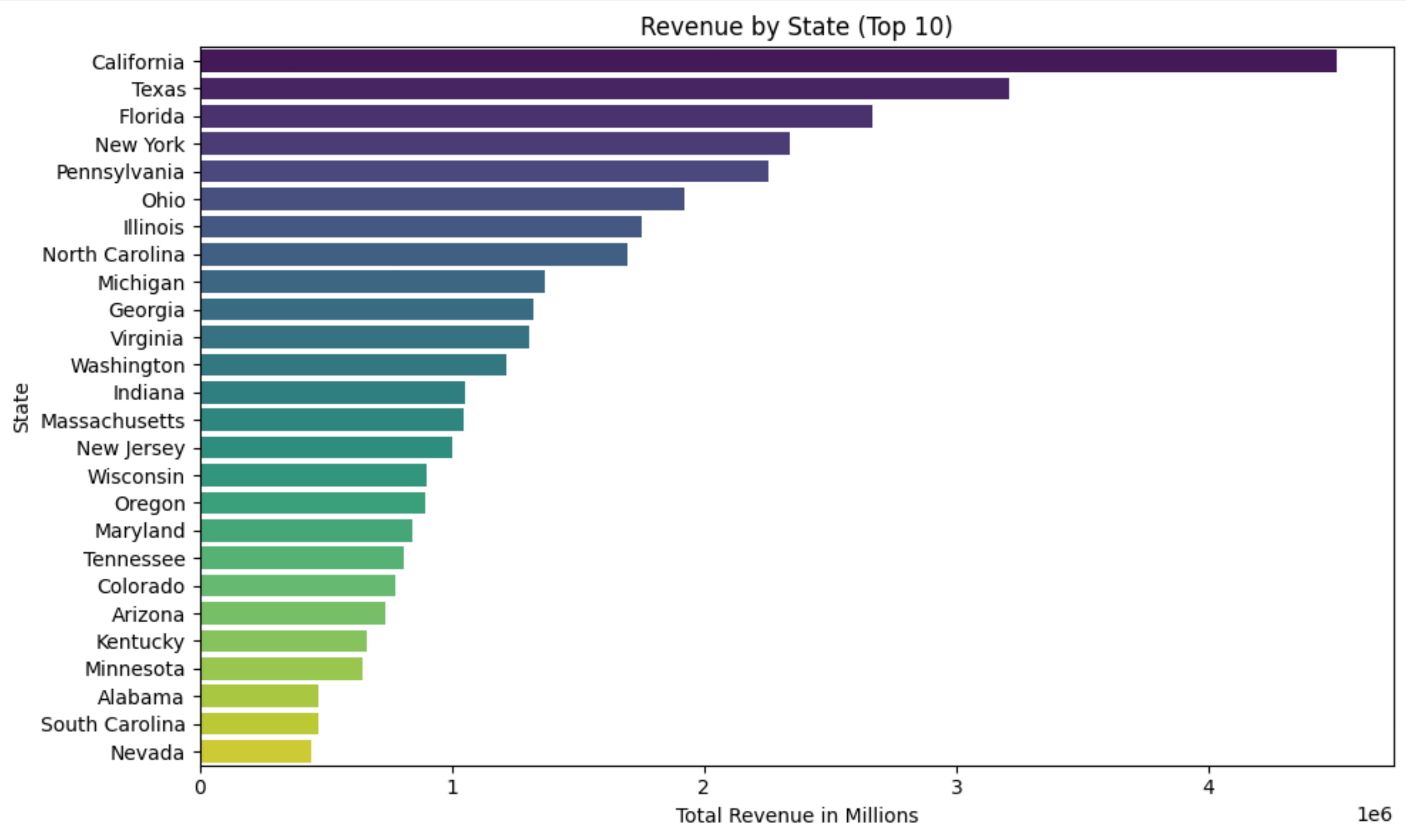
\$100,000 to \$149,999 income group generates the highest total revenue: \$9.14M
Next highest groups: \$50k–\$74,999 (\$8.26M) and \$25k–\$49,999 (\$7.34M)
Very-high-income group (\$150k+) contributes less total revenue than the upper middle income group

REVENUE BY AGE GROUP



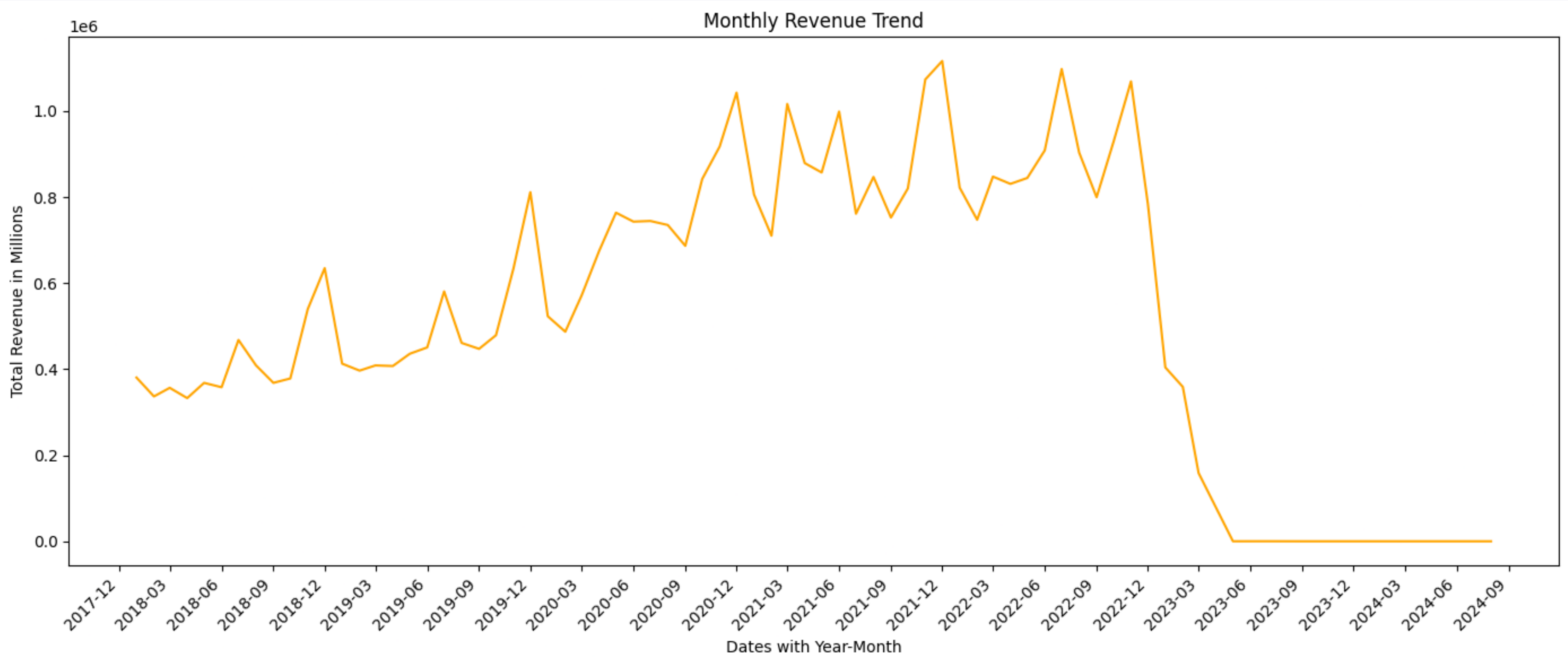
Largest age segment by revenue: 25-34 years with total revenue ≈ \$13 Million
Top two age groups 25–34 and 35–44 together account for most spending and most transactions
Older and younger groups spend less overall, consistent with fewer purchases

REVENUE BY STATE (TOP 10)



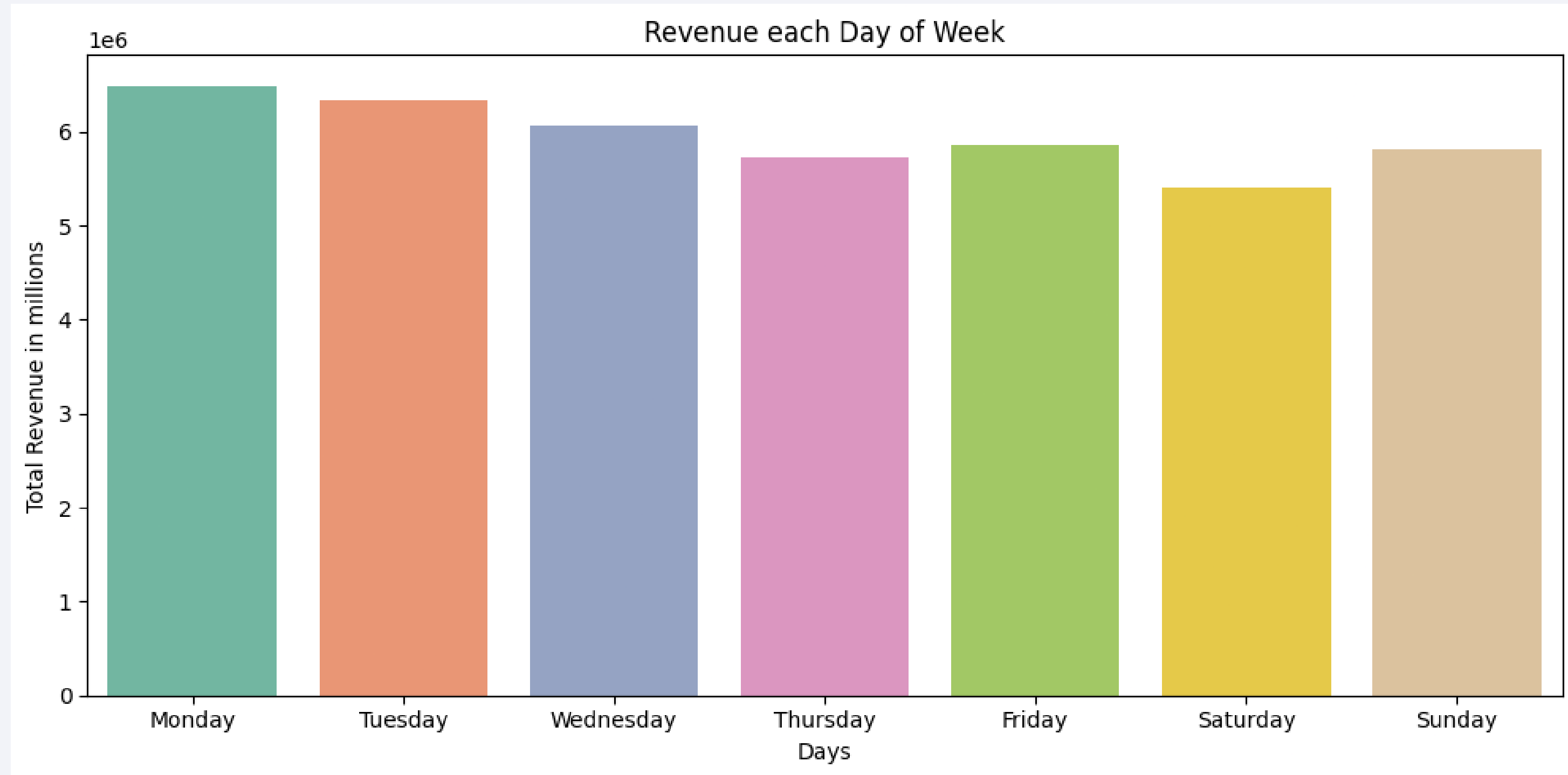
California leads with \$4.11M in revenue
Texas (\$3.08M) and Florida (\$2.53M) follow; top 5 states (CA, TX, FL, NY, PA) together exceed \$14M

TIME PATTERNS (MONTHLY TREND)



Data spans from Jan 2018 to Aug 2024
Overall monthly revenue grows until around 2022, then shows a decline around 2023
Seasonal peaks likely around typical shopping seasons (end-of-year, mid-year promotions)

TIME PATTERNS (DAY OF WEEK)

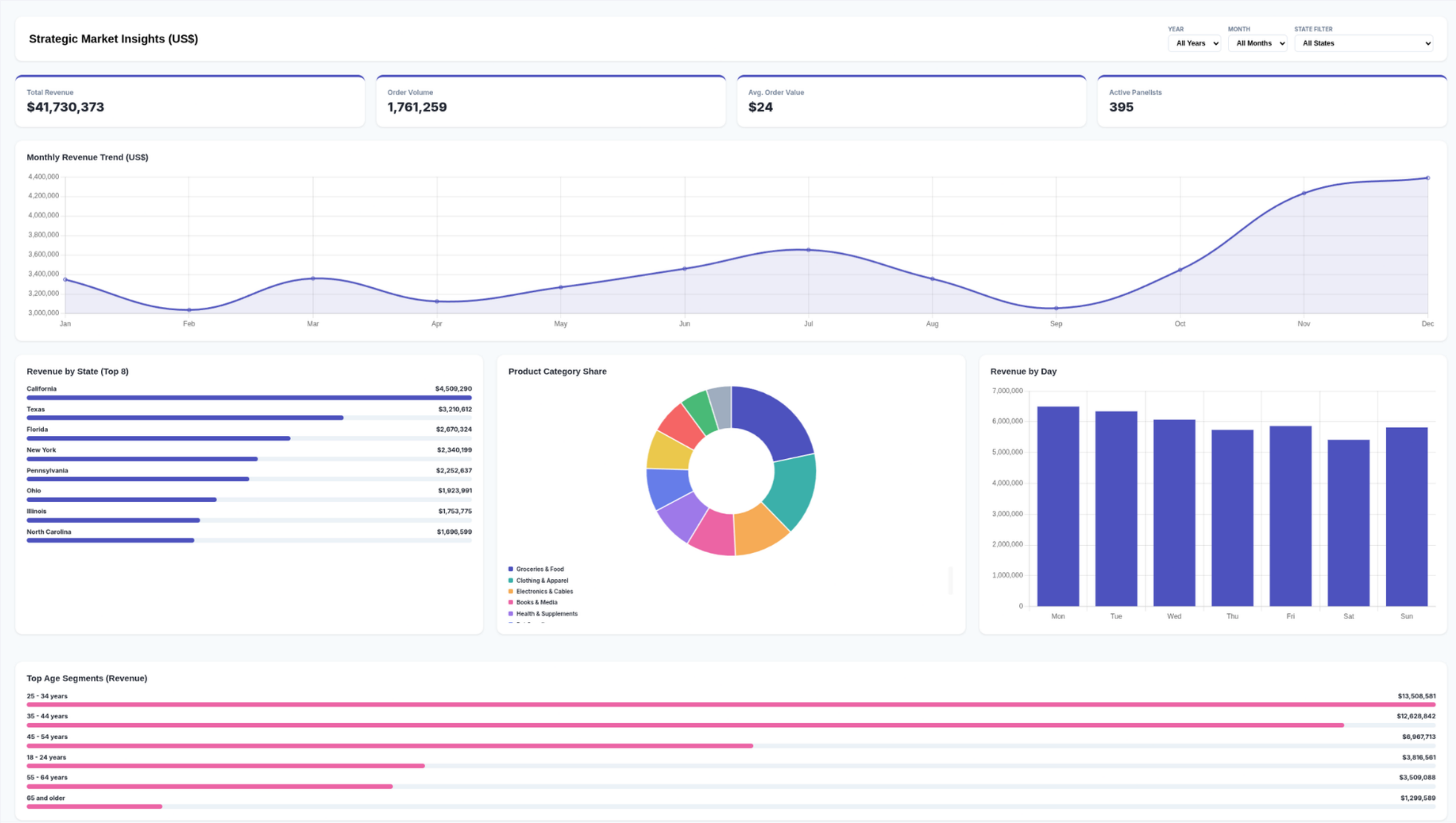


Highest revenue day: Mondays \$6 million with consistent weekly demand peak.

Lowest revenue day: Saturday with off-peak buying period

Useful for scheduling campaigns and staffing ; aim promotions at high-revenue days

INTERACTIVE DASHBOARD



Monthly Revenue Trend (US\$)



Month	Revenue (US\$)
Jan	3,350,000
Feb	3,050,000
Mar	3,350,000
Apr	3,100,000
May	3,250,000
Jun	3,450,000
Jul	3,650,000
Aug	3,350,000
Sep	3,050,000
Oct	3,450,000
Nov	4,250,000
Dec	4,350,000

Revenue by State (Top 8)

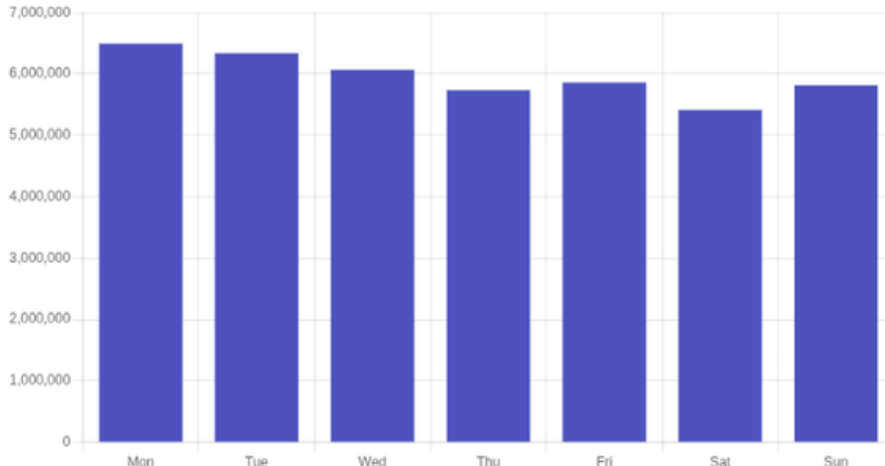
State	Revenue (US\$)
California	\$4,509,290
Texas	\$3,210,612
Florida	\$2,670,324
New York	\$2,340,199
Pennsylvania	\$2,252,637
Ohio	\$1,923,991
Illinois	\$1,753,775
North Carolina	\$1,696,599

Product Category Share



- Groceries & Food
- Clothing & Apparel
- Electronics & Cables
- Books & Media
- Health & Supplements
- ...

Revenue by Day



Day	Revenue (US\$)
Mon	6,450,000
Tue	6,300,000
Wed	6,050,000
Thu	5,750,000
Fri	5,850,000
Sat	5,400,000
Sun	5,800,000

Top Age Segments (Revenue)

Age Segment	Revenue (US\$)
25 - 34 years	\$13,508,581
35 - 44 years	\$12,628,842
45 - 54 years	\$6,967,713
18 - 24 years	\$3,816,561
55 - 64 years	\$3,509,088
65 and older	\$1,299,589

CONCLUSION – BUSINESS INSIGHTS

- **Revenue is highly concentrated:** many purchases are low value, while a small number of orders contribute disproportionately to total revenue.
- **Product demand is not evenly spread across categories:** groceries and food stand out strongly in the aggregated view, while books are highly visible in the detailed labels.
- **High-value customers** are associated with specific demographic segments, especially the income bracket around \$100,000–\$149,999 and the age group 25–34.
- **Geographic demand differs across states,** with California appearing as the strongest contributor in the current dataset.

QUESTIONS?